

Vectori și valori proprii. Diagonalizare

$$A\underline{x} = \lambda \underline{x} \quad \text{și} \quad \underline{x} \neq 0$$

$$\underline{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in K^n$$

Să spunem că $\underline{x}$ este vector propriu pt A

$$\lambda_1 \underline{x} = \lambda_2 \underline{x} \quad (\Leftrightarrow) \quad \left. \begin{array}{l} \forall i = \overline{1, n}, \lambda_1 x_i = \lambda_2 x_i \\ \exists i \text{ cu } x_i \neq 0 \end{array} \right\} \Rightarrow \lambda_1 = \lambda_2$$

Un vector propriu corespunde unei singure valori proprii.

$$\begin{pmatrix} 3 & 5 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 5 \\ 1 \end{pmatrix} = \begin{pmatrix} 20 \\ 4 \end{pmatrix} = 4 \begin{pmatrix} 5 \\ 1 \end{pmatrix}$$

$$\begin{aligned} \det \left[\lambda I_2 - \begin{pmatrix} 3 & 5 \\ 1 & -1 \end{pmatrix} \right] &= \det \left[\begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} - \begin{pmatrix} 3 & 5 \\ 1 & -1 \end{pmatrix} \right] = \\ &= \det \begin{pmatrix} \lambda - 3 & -5 \\ -1 & \lambda + 1 \end{pmatrix} = (\lambda - 3)(\lambda + 1) - 5 = \lambda^2 - 2\lambda - 8 = \\ &= (\lambda - 4)(\lambda + 2) \quad (\text{obs. răd } 4) \end{aligned}$$

λ - val proprie $\Leftrightarrow \lambda$ răd polim. caracteristic

$$S_\lambda = \left\{ \underline{x} \in K^n \mid \underline{x} \text{ este vector propriu asociat cu } \lambda \right\}$$

vectorii din K^n
(coloană)

$$\left\{ \underline{x} \in K^n \mid (\lambda I_n - A) \cdot \underline{x} = 0 \right\}$$

• $\underline{0} \in S_\lambda$ pt că sist este omogen

• $\underline{x}, \underline{y} \in S_\lambda, \alpha, \beta \in K$

$$(\lambda I_n - A)(\alpha \underline{x} + \beta \underline{y}) = \alpha \underbrace{(\lambda I_n - A)}_0 \underline{x} + \beta \underbrace{(\lambda I_n - A)}_0 \underline{y} = 0$$

Mult. sol. unui sist omogen este subspace.

$$P \cdot D = P^{-1} \cdot A \cdot P \quad / P^{-1}$$

$$P \cdot D \cdot P^{-1} = A$$

$$D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n) = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}$$

← aici nu e obligatoriu ca valorile să fie distincte

$$(\underline{x}_1 \ \underline{x}_2 \ \dots \ \underline{x}_n) \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix} = A (\underline{x}_1, \underline{x}_2 \ \dots \ \underline{x}_n)$$

$$(\lambda_1 \underline{x}_1, \lambda_2 \underline{x}_2, \dots, \lambda_n \underline{x}_n) = (A \underline{x}_1, A \underline{x}_2, \dots, A \underline{x}_n)$$

$$\lambda_i \underline{x}_i = A \underline{x}_i, \quad 1 \leq i \leq n$$